

SOME CHARACTERIZATIONS OF NORMAL AND PERFECTLY NORMAL SPACES

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We present here first a theorem (Theorem 1) on lattices; then with the aid of this theorem, we prove the characterization theorems (Theorems 2 and 3) for normal and perfectly normal spaces (a normal space is said to be perfectly normal if every closed set in it is the intersection of denumerably many open sets; this definition was given by Čech [2; 40]). Special cases of Theorem 1 for sets and for functions are due to W. Sierpiński, H. Hahn and F. Hausdorff (compare [4] and [3]). For metric spaces, Theorem 2, property (2) is due to Hahn [3; 292], and (3) was first proved by H. Tietze [5]. For Euclidean n -space, Theorem 3, (2) is a classical result of R. Baire [1; 124]; it was generalized to metric spaces by Tietze [5]; (3) of Theorem 3 (for metric spaces) is a theorem of Tietze [5].

Let L be a lattice which can be embedded in a lattice M such that: (1) the l.u.b. and g.l.b. of every denumerable set in L exist in M , and (2) $s \in M$ and $t_n \in L$ imply g.l.b. $\{s \cup t_n\}$ and l.u.b. $\{s \cap t_n\} \in M$ with $s \cup \text{g.l.b.}\{t_n\} = \text{g.l.b.}\{s \cup t_n\}$ and $s \cap \text{l.u.b.}\{t_n\} = \text{l.u.b.}\{s \cap t_n\}$. Let L_σ be the set of l.u.b.'s of denumerably many elements in L (L_δ be the set of g.l.b.'s). We now prove

THEOREM 1. *If $s \in L_\delta$, $t \in L_\sigma$, and $s \subseteq t$, then there is a $u \in L_\sigma L_\delta$, and $s \subseteq u \subseteq t$.*

Proof. Let $s = \text{g.l.b.}\{s_n\}$, and $t = \text{l.u.b.}\{t_n\}$. Clearly we may assume $s_n \supseteq s_{n+1}$ and $t_n \subseteq t_{n+1}$ for every n . Let $u_n = (s_1 \cap t_1) \cup (s_2 \cap t_2) \cup \cdots \cup (s_n \cap t_n)$, and $v_n = u_{n-1} \cup s_n$. Let l.u.b. $\{u_n\} = u$ and g.l.b. $\{v_n\} = v$. Now it is readily seen that $v_n \subseteq u \cup s_n$ for every n ; hence, g.l.b. $\{v_n\} \subseteq \text{g.l.b.}\{u \cup s_n\}$ or $v \subseteq u \cup \text{g.l.b.}\{s_n\} = u \cup s$. Since $v_n \supseteq s_n$, therefore, g.l.b. $\{v_n\} \supseteq \text{g.l.b.}\{s_n\}$ or $v \supseteq s$. Consequently $v \cup s = v \subseteq u \cup s$. The last statement implies that $v = u \cup s$ since $v \supseteq u$ (this can be seen as follows: $v_n \supseteq u_{n+p}$ for every n and p , hence $v_n \supseteq \text{l.u.b.}\{u_{n+p}\}$; finally we have g.l.b. $\{v_n\} \supseteq u$, or $v \supseteq u$). We next show that $u \supseteq s$. For we have $u = \text{l.u.b.}\{s_n \cap t_n\} \supseteq \text{l.u.b.}\{s \cap t_n\} = s \cap \text{l.u.b.}\{t_n\} = s \cap t = s$, since $s \subseteq t$ by hypothesis. Thus $v = u \cup s = u$. So $u \in L_\sigma L_\delta$, and clearly $s \subseteq u \subseteq t$.

In the sequel, unless stated to the contrary, every function is assumed to be real-valued and defined over a topological space R , and $f \leq g$ means $f(x) \leq g(x)$ for every x in R .

THEOREM 2. *The following properties about a topological space R are equivalent:*

- (1) R is normal.

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(2) If f and g are respectively upper and lower semi-continuous functions such that $f \leq g$, then there is a continuous function h with the property that $f \leq h \leq g$.

(3) If ϕ is bounded from below and continuous over a closed set A , then there is an h which is continuous such that $h(x) = \phi(x)$ for $x \in A$, and $h \leq \phi$.

Proof. We show that (1) $\rightarrow$ (2) $\rightarrow$ (3) $\rightarrow$ (1).

(1) $\rightarrow$ (2): Since $\phi = \Phi/(1 + |\Phi|)$ (thus $\Phi = \phi/(1 - |\phi|)$) defines an order preserving homeomorphism of the real line onto the interval $(-1, 1)$, it is readily seen that without loss of generality we may assume that $-1 < f(x) \leq g(x) < 1$ (for every x). Furthermore, since $\phi = (1/2)(1 + \Phi)$ defines an order preserving homeomorphism of $(-1, 1)$ onto $(0, 1)$, we may restrict ourselves to the case where $0 < f(x) \leq g(x) < 1$.

Let m and n be positive integers with $m < n$. Let $A_{mn} = R(x : g(x) \leq m/n)$ and $B_{mn} = R(x : f(x) \geq m/n + 1/2n)$. Now A_{mn} and B_{mn} are closed (since f is upper semi-continuous and g is lower semi-continuous) and mutually exclusive. According to Urysohn's lemma, there is a continuous function f_{mn} such that $f_{mn}(x) = (m/n) + (1/2n)$ for $x \in A_{mn}$, $f_{mn}(x) = 1$ for $x \in B_{mn}$, and $f_{mn} \geq (m/n) + (1/2n)$. Let $f_n(x) = \min(f_{1n}(x), f_{2n}(x), \dots, f_{n-1,n}(x))$. It is not difficult to verify that for any x either $f_n(x) < g(x)$ or $|f_n(x) - g(x)| < 3/2n$; and $f_n \geq f$ (since $f_{mn} \geq f$, as is seen from the above construction). (In this paragraph, we are using C. H. Dowker's simplification of our original proof.)

Thus $f \leq \text{g.l.b.}\{f_n\} \leq g$. Now $-g \leq -\text{g.l.b.}\{f_n\}$, with $-g$ upper semi-continuous and $-\text{g.l.b.}\{f_n\}$ lower semi-continuous. According to the above argument, there is a sequence $\{g_n\}$ of continuous functions such that $-g \leq \text{g.l.b.}\{g_n\} \leq -\text{g.l.b.}\{f_n\}$, or $f \leq \text{g.l.b.}\{f_n\} \leq \text{l.u.b.}\{-g_n\} \leq g$.

Hence according to Theorem 1, there is a function h such that it is both the l.u.b. and g.l.b. of sequences of continuous functions, and $\text{g.l.b.}\{f_n\} \leq h \leq \text{l.u.b.}\{-g_n\}$ (the lattice L in this case is the set of all finite-valued continuous functions; M is the set of all functions which may assume the values $-\infty$ and ∞ as well as finite values; $f \cup g = \max(f, g)$ and $f \cap g = \min(f, g)$ with l.u.b. and g.l.b. having the usual meanings for functions). Thus h is both lower and upper semi-continuous, i.e. h is continuous. Evidently $f \leq h \leq g$.

(2) $\rightarrow$ (3): Let $g(x) = \text{l.u.b.}_{G \in \mathfrak{D}} \{\text{g.l.b.}_{x \in G} \phi(x)\}$ (where G is an open set containing x , and $\mathfrak{D}$ is the collection of all open sets in R). Now g is lower semi-continuous, and $g(x) = \phi(x)$ for $x \in A$ and $g \leq \phi$. Let $f(x) = \phi(x)$ for $x \in A$, and $f(x) = \lambda$ for $x \in R - A$ (λ is a lower bound of $\phi(x)$). Then $f \leq g$, and f is upper semi-continuous. Thus (2) implies there is a continuous h such that $f \leq h \leq g$. It is clear that $h \leq \phi$, and $h(x) = \phi(x)$ for $x \in A$.

(3) $\rightarrow$ (1): Let A and B be two disjoint closed sets in R . Let $\phi(x) = 1$ for $x \in A$ and $\phi(x) = 0$ for $x \in R - A$. Then $\phi(x)$ is bounded from below and continuous over the closed set $A + B$. Thus there is a continuous h such that $h(x) = \phi(x)$ for $x \in A + B$. Let $G_1 = R(x : h(x) > 1/2)$, $G_2 = R(x : h(x) < 1/2)$. Since $h(x)$ is continuous, G_1 and G_2 are disjoint open sets covering A and B respectively; thus R is normal.

COROLLARY. (*Tietze-Urysohn extension theorem*) Let $\phi(x)$ be a function defined and continuous over a closed set A of a normal space R , then there is a function h continuous over R such that $h(x) = \phi(x)$ for $x \in A$.

Proof. We first consider the case where $\phi(x)$ is bounded. Let $\text{l.u.b.}_{x \in A} \phi(x) = \eta$ and $\text{g.l.b.}_{x \in A} \phi(x) = \lambda$. Let $f(x) = \phi(x)$ for $x \in A$ and $f(x) = \lambda$ for $x \in R - A$; $g(x) = \phi(x)$ for $x \in A$ and $g(x) = \eta$ for $x \in R - A$. Then f and g satisfy the hypothesis of (2); thus there is a continuous h such that $f \leq h \leq g$. The function h has the properties required by the corollary. We turn now to the unbounded case. The function $\Phi = \phi/(1 + |\phi|)$ has as bounds -1 and 1 ; these values are never assumed. Let H be a continuous function such that $H(x) = \Phi(x)$ for $x \in A$ and $-1 \leq H(x) \leq 1$ for $x \in R$ (such an H exists according to the previous case). Let $B = R(x : |H(x)| = 1)$. A and B are disjoint closed sets. Let $f(x)$ be continuous such that $f(x) = 1$ for $x \in A$, $f(x) = 0$ for $x \in B$, and $0 \leq f(x) \leq 1$ for every x . The function $h(x) = H(x)f(x)/(1 - |H(x)f(x)|)$ is then a required extension of $\phi(x)$.

THEOREM 3. The following conditions about a topological space R are equivalent:

- (1) R is perfectly normal.
- (2) Every lower semi-continuous function is the limit of a monotonically increasing sequence of continuous functions.
- (3) Every lower semi-continuous function $\phi(x)$, which is continuous in a closed set A , is the limit of a monotonically increasing sequence of continuous functions which have the same values as $\phi(x)$ for $x \in A$.

Proof. (1) $\rightarrow$ (2). We first prove that if a lower semi-continuous function f assumes only two values, then it is the limit of a monotonically increasing sequence of continuous functions. Let the two values be α and β . Evidently we only need to prove the case where $\alpha = 1$ and $\beta = 0$. Since the function is lower semi-continuous, $A = R(x : f(x) = 0)$ is closed. Because R is perfectly normal $A = \bigcap_{n=1}^{\infty} G_n$, where G_n is an open set and $G_n \supseteq G_{n+1}$. Let f_n be a continuous function with $f_n(x) = 1$ for $x \in R - G_n$, $f_n(x) = 0$ for $x \in A$, and $1 \geq f_n \geq 0$. Clearly $h(x) = \sum_{n=1}^{\infty} 2^{-n} f_n(x)$ is continuous, and $A = R(x : h(x) = 0)$. Let $g_n(x) = \min(1, nh(x))$. Then $\{g_n\}$ is a monotonically increasing sequence of continuous functions with f as limit.

Now suppose f assumes n values, say $\alpha_1 > \alpha_2 > \dots > \alpha_n$. Let $\phi_i(x) = \alpha_{i+1}$ for $x \in R(x : f(x) \leq \alpha_{i+1})$, and $\phi_i(x) = \alpha_1$ for $x \in R - R(x : f(x) \leq \alpha_{i+1})$; $1 \leq i \leq n - 1$. Since $R(x : f(x) \leq \alpha_{i+1})$ is a closed set, ϕ_i is lower semi-continuous. Now $f = \min(\phi_1, \phi_2, \dots, \phi_{n-1})$. Let $\{\phi_{ij}\}$ be a monotonically increasing sequence of continuous functions having ϕ_j as its limit. Then $\{\min(\phi_{i1}, \phi_{i2}, \dots, \phi_{i,n-1})\}$ is a sequence with f as its limit.

We proceed now to the general case. As in the above, we may restrict ourselves to the case where $0 < f(x) < 1$. We must, in order to include the situation where the lower semi-continuous function is unbounded from below, produce a sequence $\{f_n\}$ such that $f_n(x) > 0$ for every x . Let $A_{mp} = R(x : (p + 1)/m \geq f(x) > p/m)$;

m and p are integers with $0 \leq p \leq m-1$. For every fixed m , let $g_m(x) = p/m$ for $x \in A_{mp}$. It is clear that $g_m(x)$ is lower semi-continuous. Thus there is a monotonically increasing sequence $\{g_{mi}\}$ such that $\lim g_{mi} = g_m$. We note that $g_m \leq f$, and $g_{mi}(x)$ may be assumed to be ≥ 0 for every x . Let the double sequence $\{g_{mi}\}$ be arranged as a single sequence $\{\psi_n\}$, and let $\phi_n = \max(\psi_1, \dots, \psi_n)$. Since $f(x) > 0$, $\phi = \sum_{i=1}^{\infty} 2^{-i} \phi_i(x) > 0$ for every x . It is readily seen that $\phi \leq f$. Let $f_n = \max(\phi, \phi_n)$; then $\{f_n\}$ is a monotonically increasing sequence of continuous functions having f as limit and never assuming the value 0.

(1) $\rightarrow$ (3). We again restrict ourselves to the case where $0 < \phi(x) < 1$; and in order to include the case where the lower semi-continuous function is unbounded from below, we shall produce a sequence $\{\phi_n\}$ such that $\phi_n(x) > 0$ for every x and $\phi_n(x) = \phi(x)$ for $x \in A$ (and the ϕ_n 's are continuous with $\lim \phi_n = \phi$). Now according to the above paragraph, there is a monotonically increasing sequence $\{f_n\}$ of continuous functions such that $\lim f_n = \phi$ and $f_n(x) > 0$ for every x . Now since R is normal, according to (3) of Theorem 2, there is a function h continuous over R with $h(x) = \phi(x)$ for $x \in A$ and $h \leq \phi$. Thus $\{\phi_n\}$ (where $\phi_n = \max(h, f_n)$) is a sequence with the required properties.

(3) $\rightarrow$ (2). This is manifest.

(2) $\rightarrow$ (1). Let A be a closed set. Let $f(x) = 0$ for $x \in A$ and $f(x) = 1$ for $x \in R - A$. Then f is the limit of a monotonically increasing sequence of continuous functions $\{f_n\}$. Let $\phi_n = \max(0, f_n)$. Let $\phi = \sum_{i=1}^{\infty} (1/2^i) \phi_i$. Then $\phi(x) = 0$ if and only if $x \in A$ and $\phi(x) > 0$ elsewhere. Thus $A = \bigcap_{n=1}^{\infty} G_n$, where $G_n = R(x : \phi(x) < 1/n)$. This shows that every closed set in R is a G_δ . Let A and B be disjoint closed sets in R . Let $\phi_i(x)$ ($i = 1, 2$) be continuous functions such that $\phi_i(x) \geq 0$ and $A = R(x : \phi_1(x) = 0)$, $B = R(x : \phi_2(x) = 0)$. Then $\phi_1 - \phi_2$ is positive on B and negative on A . Let $G_1 = R(x : \phi_1(x) - \phi_2(x) < 0)$, $G_2 = R(x : \phi_1(x) - \phi_2(x) > 0)$. Then G_1 and G_2 are disjoint open sets covering A and B respectively. Thus R is normal. Since R is normal and every closed set in R is a G_δ , R is perfectly normal.

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